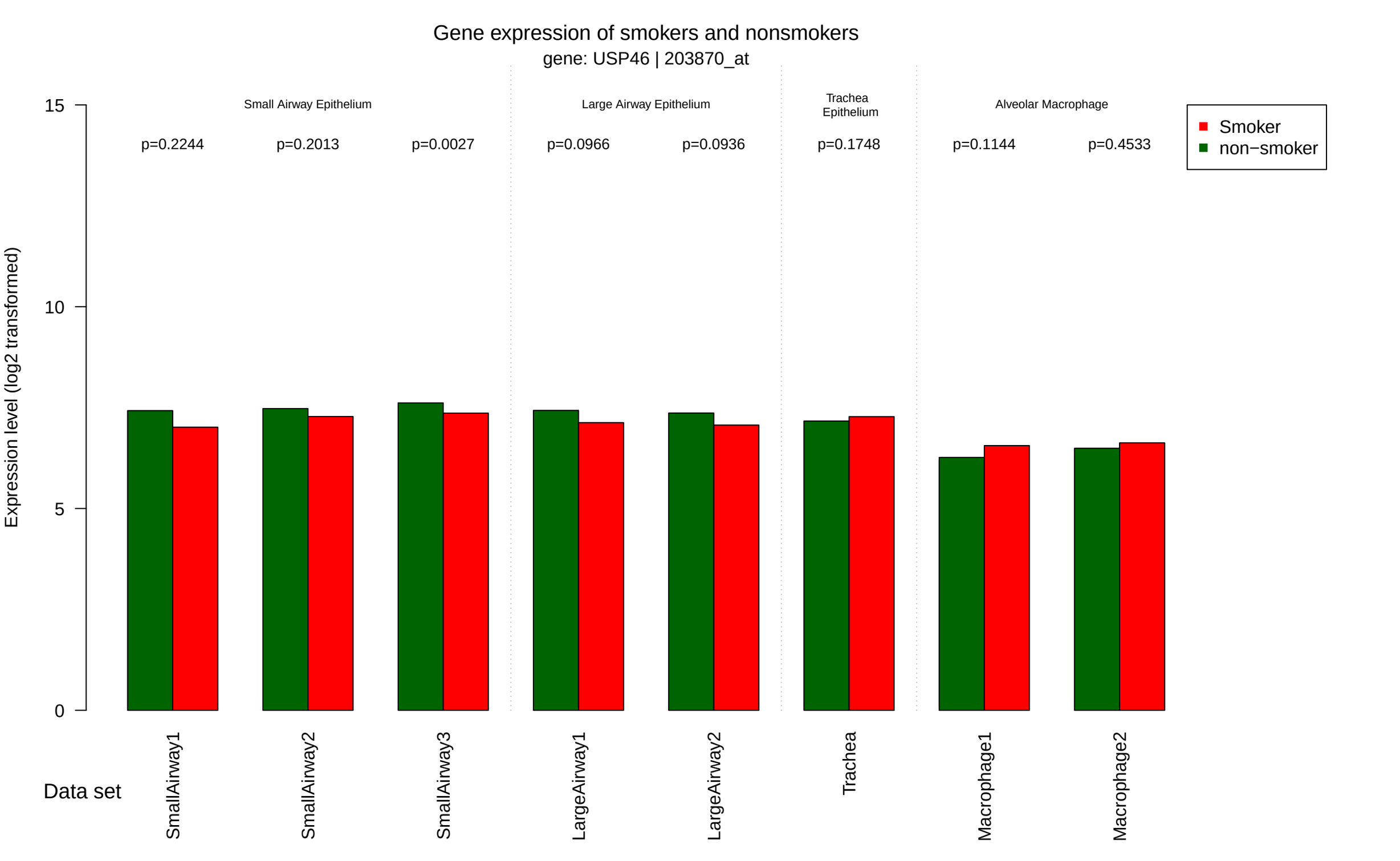
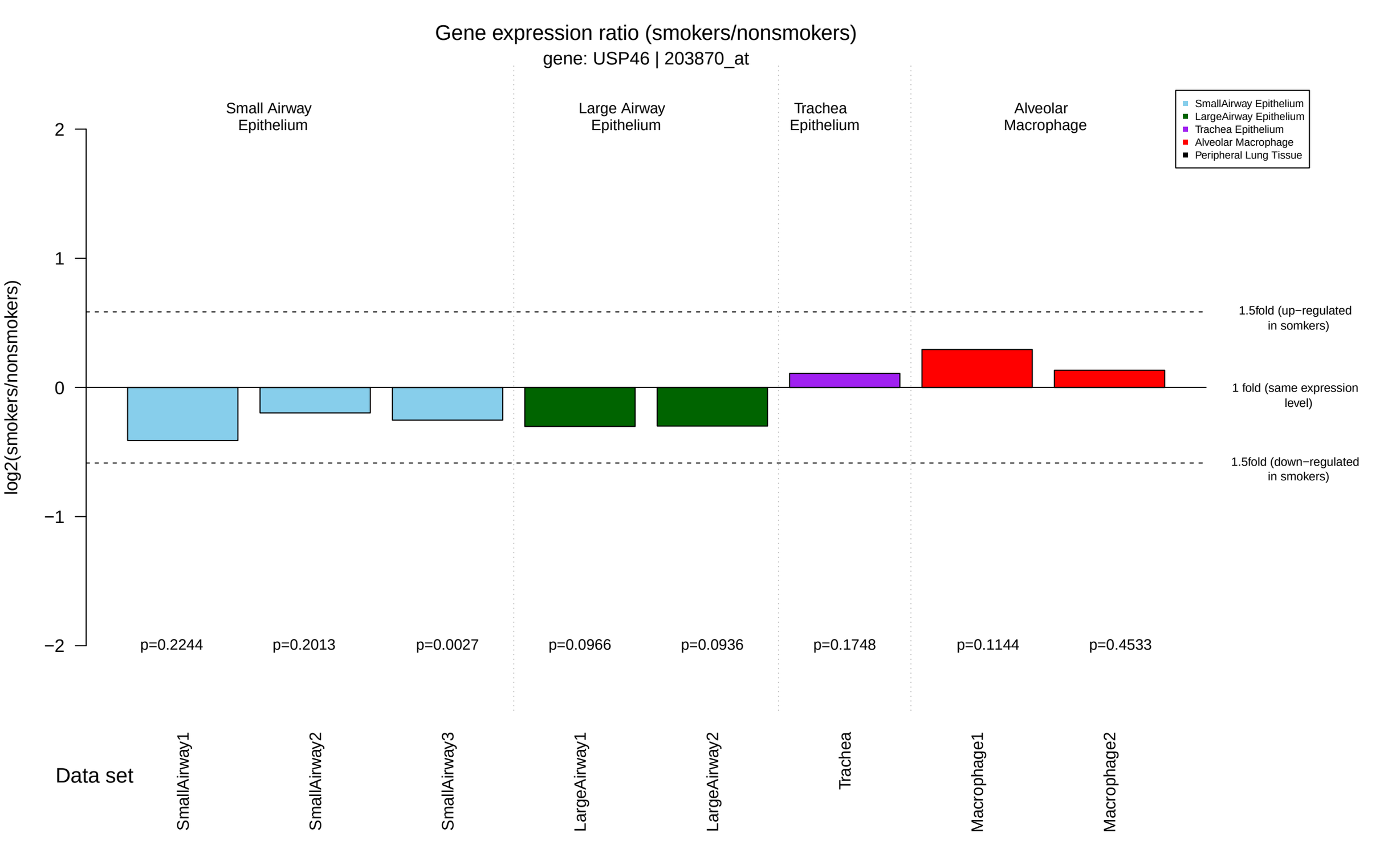
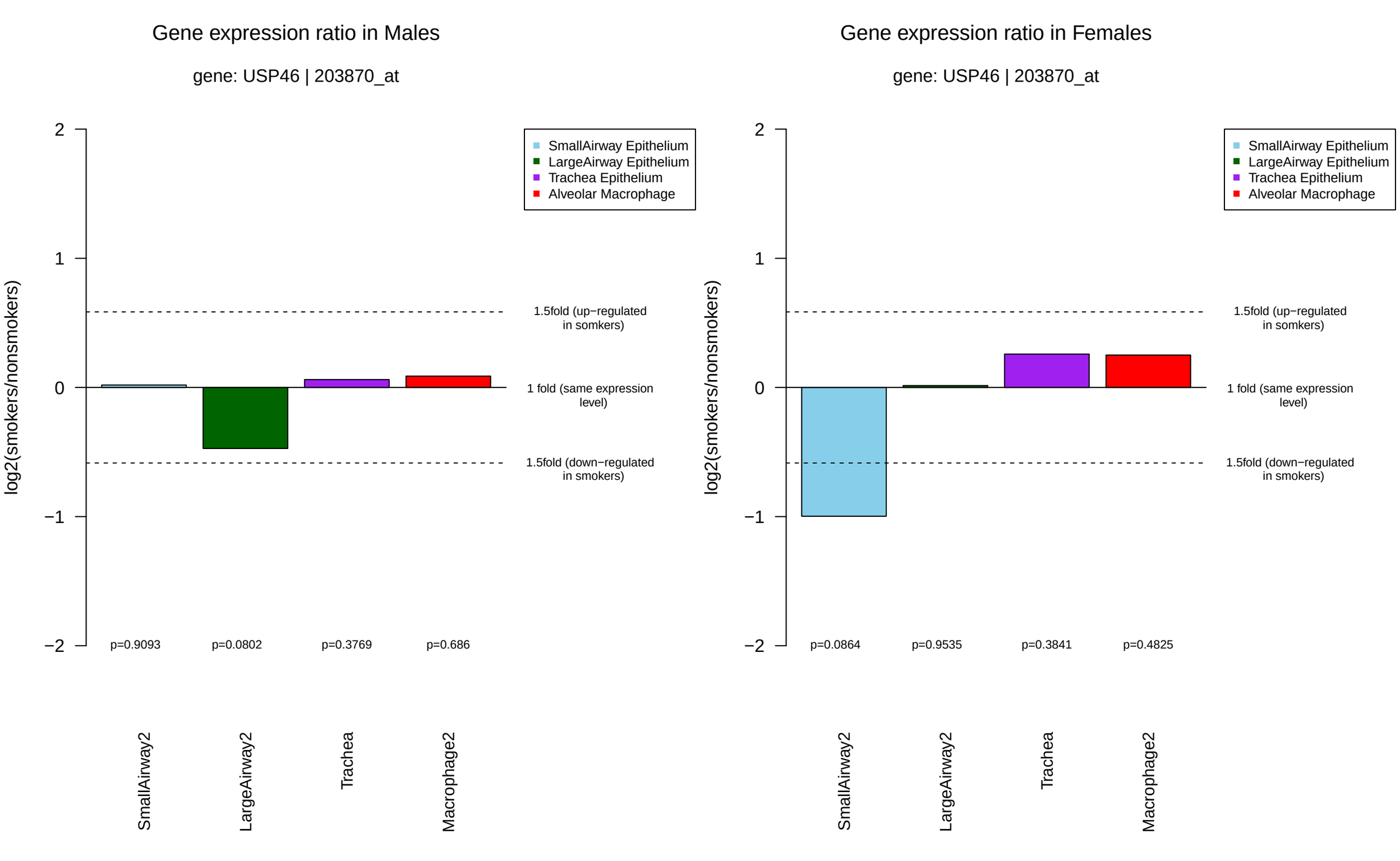
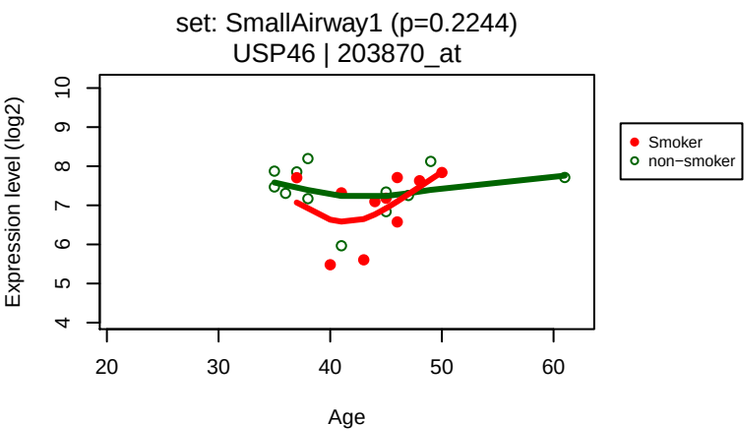
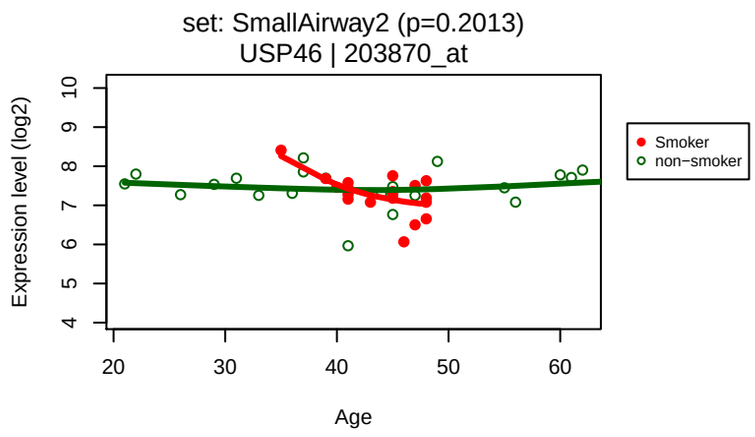
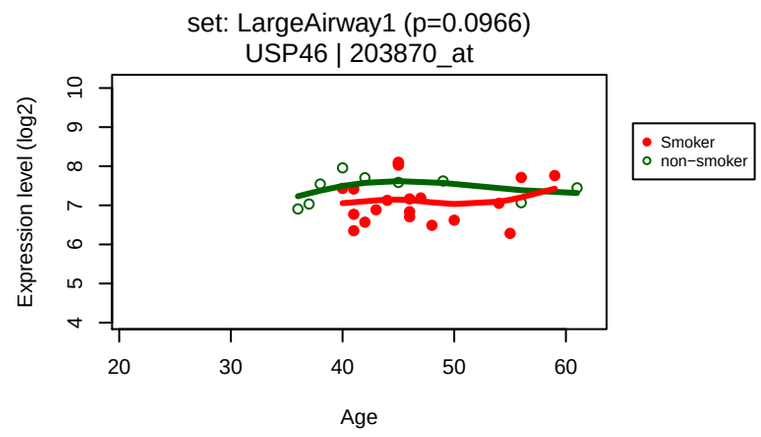

Gene expression of smokers and nonsmokers
gene: USP46 | 203870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2244
p=0.2013
p=0.0027
p=0.0966
p=0.0936
p=0.1748
p=0.1144
p=0.4533
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

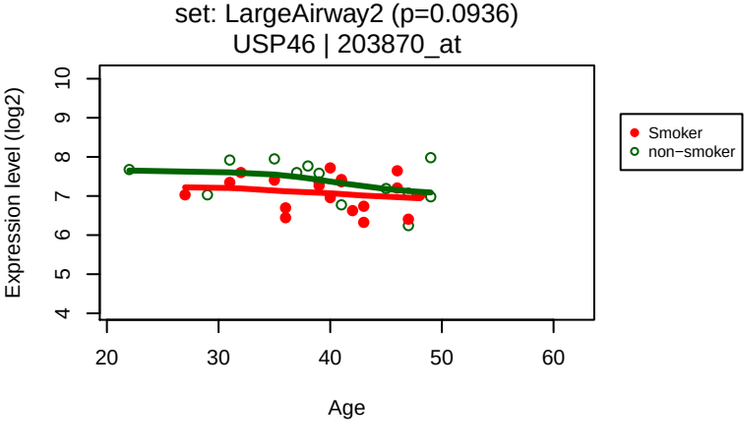

Gene expression ratio (smokers/nonsmokers)
gene: USP46 | 203870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2244
p=0.2013
p=0.0027
p=0.0966
p=0.0936
p=0.1748
p=0.1144
p=0.4533
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

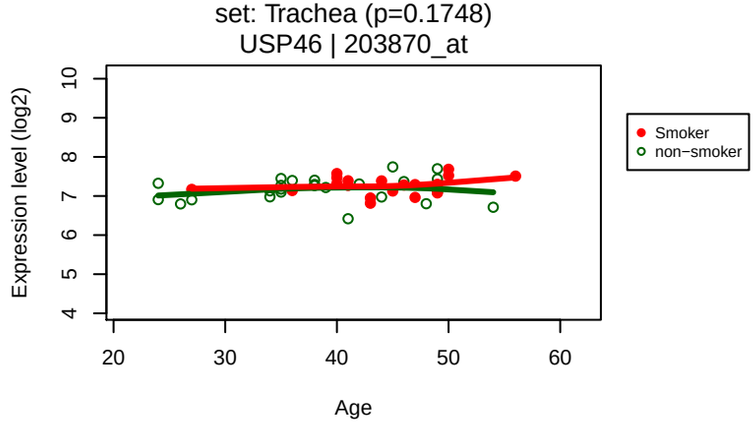

Gene expression ratio in Males
gene: USP46 | 203870_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9093
p=0.0802
p=0.3769
p=0.686
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP46 | 203870_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0864
p=0.9535
p=0.3841
p=0.4825
SmallAirway2
LargeAirway2
Trachea
Macrophage2

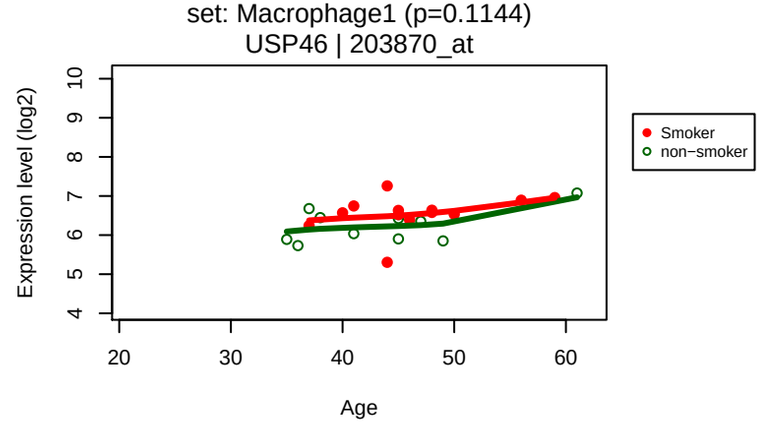

set: SmallAirway1 (p=0.2244)
USP46 | 203870_at
Expression level (log2)
Age
Smoker
non−smoker

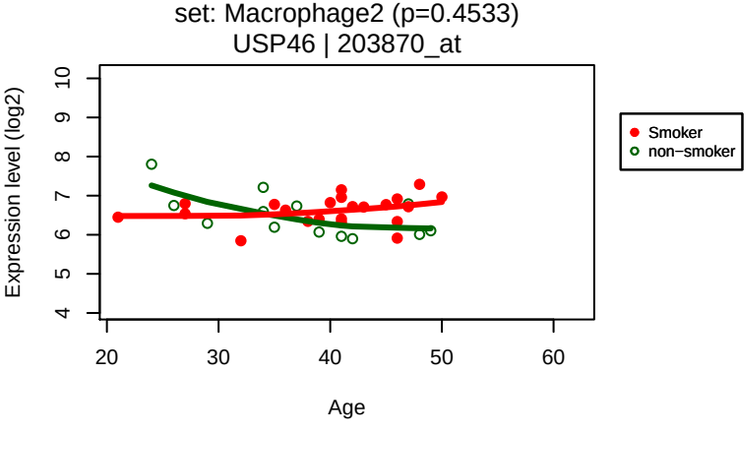

set: SmallAirway2 (p=0.2013)
USP46 | 203870_at
Expression level (log2)
Age
Smoker
non−smoker

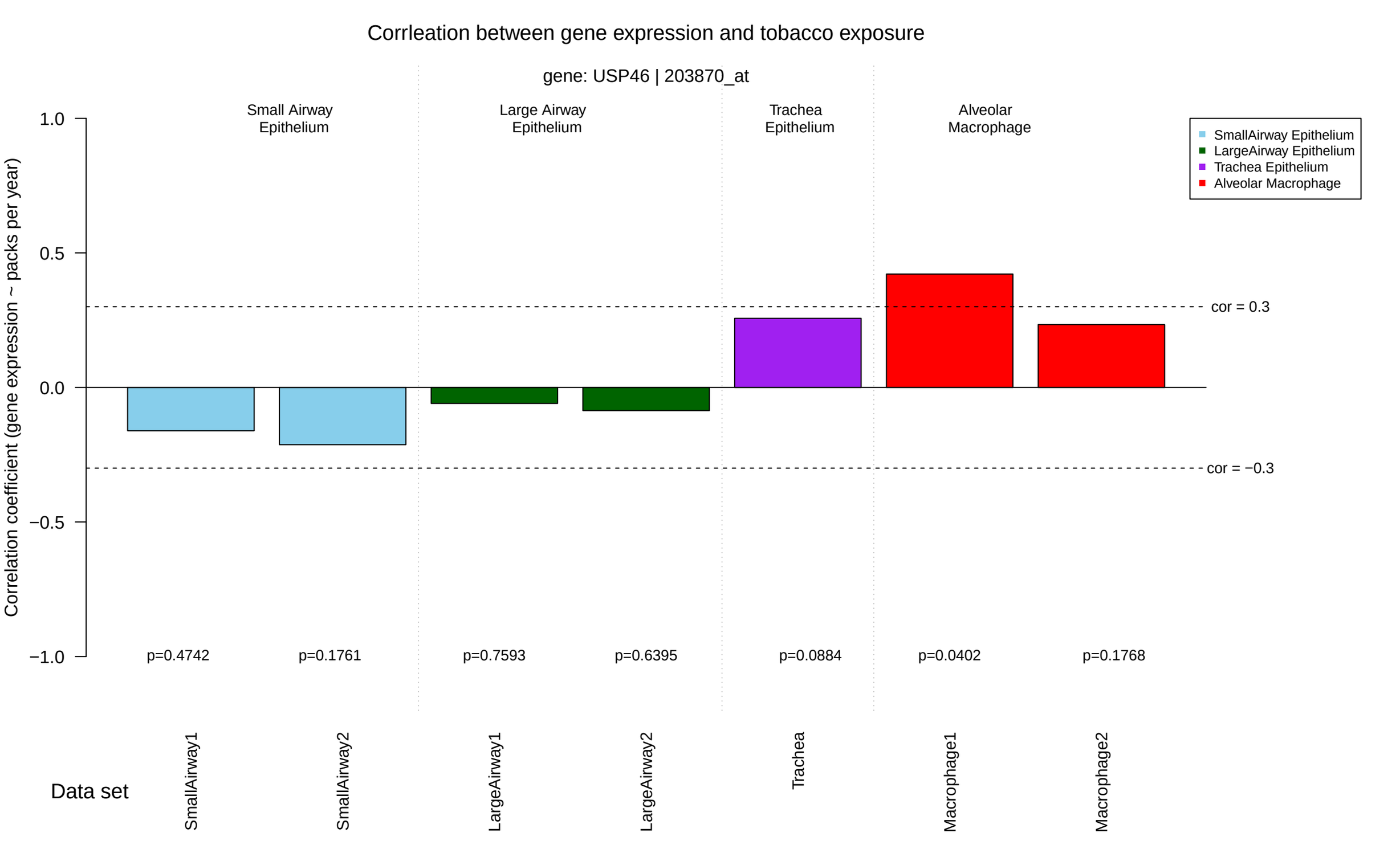

set: LargeAirway1 (p=0.0966)
USP46 | 203870_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0936)
USP46 | 203870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1748)
USP46 | 203870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1144)
USP46 | 203870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4533)
USP46 | 203870_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP46 | 203870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4742
p=0.1761
p=0.7593
p=0.6395
p=0.0884
p=0.0402
p=0.1768
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2